Pratham Bhavin Patel

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SKILLS:

- Languages: Python, Java, JavaScript, C++, C#, SQL, Bash, HTML
- Backend & Infrastructure: FastAPI, Python Data Platforms, Spring Boot, Node.js, Distributed Systems, Microservices, Common Design Patterns
- Cloud & DevOps: AWS Certified Cloud Practitioner, AWS (EC2), Docker, Git, REST APIs, Continuous Integration & Deployment (CI/CD), AI Development Tools (GitHub Copilot, Claude)
- Databases: PostgreSQL, MySQL, TimescaleDB, NoSQL
- Engineering Principles: Systems Engineering, Process Review & Optimization, Infrastructure Analytics, Automation Pipelines, Object-Oriented Design (OOD)

EDUCATION:

Texas A & M University | M.S. Data Science | College Station, TX **Aug 2026 – May 2028**

Gujarat Technological University| B.Tech Information Technology| Ahmedabad, India **Apr 2019 – Apr 2023**

EXPERIENCE:

Software Developer / Tester - Texas A&M University, Technology Services **Jul 2026 – Present**

- Designed and implemented a permission-aware tracking and data classification service from scratch, building recursive unit-hierarchy traversal in C# to power secure corporate workflows across 5 organizational branches.
- Executed an extensive infrastructure and process review of legacy data logic, eliminating N+1 database queries to optimize server resource utilization and cut processing latency from 10 minutes to under 30 seconds for 10K+ users.
- Applied strict engineering principles to drive infrastructure scalability and technology enhancements, collaborating with senior engineers to deploy 45+ core production system patches across 5 application servers.
- Partnered with product leads to restructure feature architecture, extending a Blazor UI with 6 high-throughput bulk-action workflows to deliver clean process improvements and reduce application failure loops to zero.

Software Engineering Intern – Bisag (Govt. of India) **Nov 2022 – May 2023**

- Designed automated Python-based data platform architectures using Python and SQL to update, version, and evaluate performance data anomalies across 45,000+ continuous telemetry samples.
- Collaborated with validation teams to build internal data pipelines and tool orchestration layers, testing and debugging high-throughput streams to systematically improve product data reliability.
- Architected and shipped an end-to-end full-stack data service via FastAPI and Python on AWS EC2, building performant RESTful APIs benchmarked to sustain 1,500+ requests per second to provide reliable, and actionable data.

Research Assistant / Student Developer - Adani University **Nov 2021 – Oct 2022**

- Turned ambiguous requirements into shipped software by designing and writing custom Python data collection modules from scratch, optimizing node device data processing efficiency by 35%.
- Owned technical choices end-to-end to build a reliable Python network data layer, establishing custom data synchronization logic that reduced system-level packet drop rates by 22% across a centralized database cluster.
- Collaborated across a multi-disciplinary team to troubleshoot, test, and safely deploy the automated Python data ingestion network across ~4,000 concurrent active validation nodes under real-world conditions.

PROJECTS:

High-Throughput Financial Simulation & Metrics Engine (Python, TimescaleDB, Redis, Docker)

- Defined and led the end-to-end development of a Python-based data platform leveraging TimescaleDB to ingest 5M+ daily product performance data points across containerized environments.
- Constructed an automated data pipeline using Python to model anomalous performance events, building analytics dashboards to empower validation teams to extract actionable, data-driven insights.

Autonomous Scenario & Core Event Matching Engine (C++)

- Built an end-to-end high-performance systems simulation network engine from scratch using solid data structures and algorithmic fundamentals (std::set, unordered_map) to execute complex concurrent commands under strict sub-millisecond constraints.
- Implemented custom priority comparators to handle strict real-time event matching, state tracking, and sequential notifications across highly concurrent, business-critical data streams.